ALRYADA UNIVERSITY
FOR SCIENCE AND TECHNOLGY

FACULTY OF NURSING

LECTURE 7

ANTIMICROBIAL RESISTANCE

By
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Intended Learning Outcomes (ILOs)

By the end of this lecture, the student will be able to:

1. List the main mechanisms of bacterial drug resistance.

2. Differentiate between genetic and non-genetic resistance.

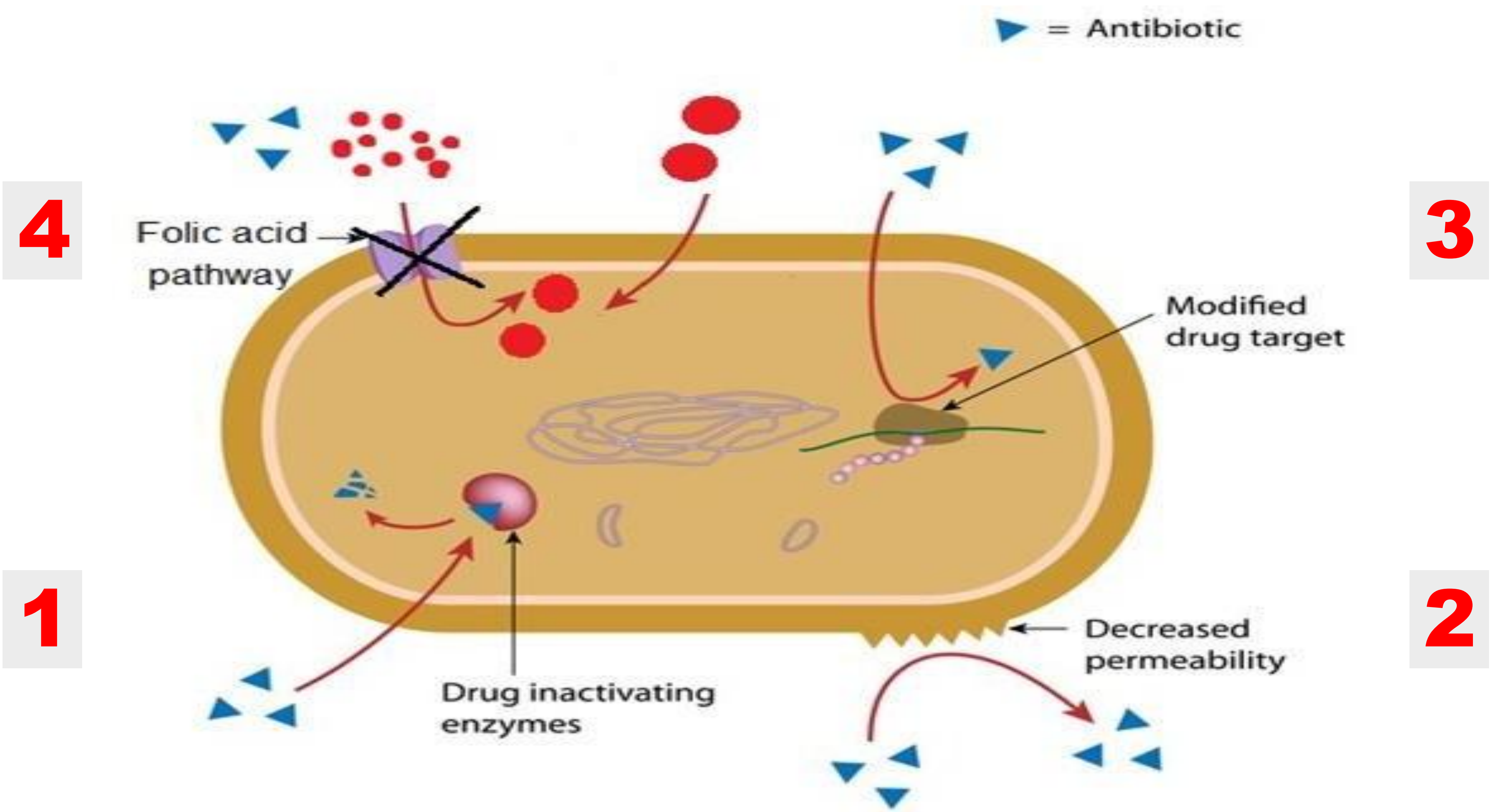
3. Recognize how bacteria inactivate drugs, alter targets, and reduce drug entry.

4. Identify methods of antibiotic susceptibility testing .

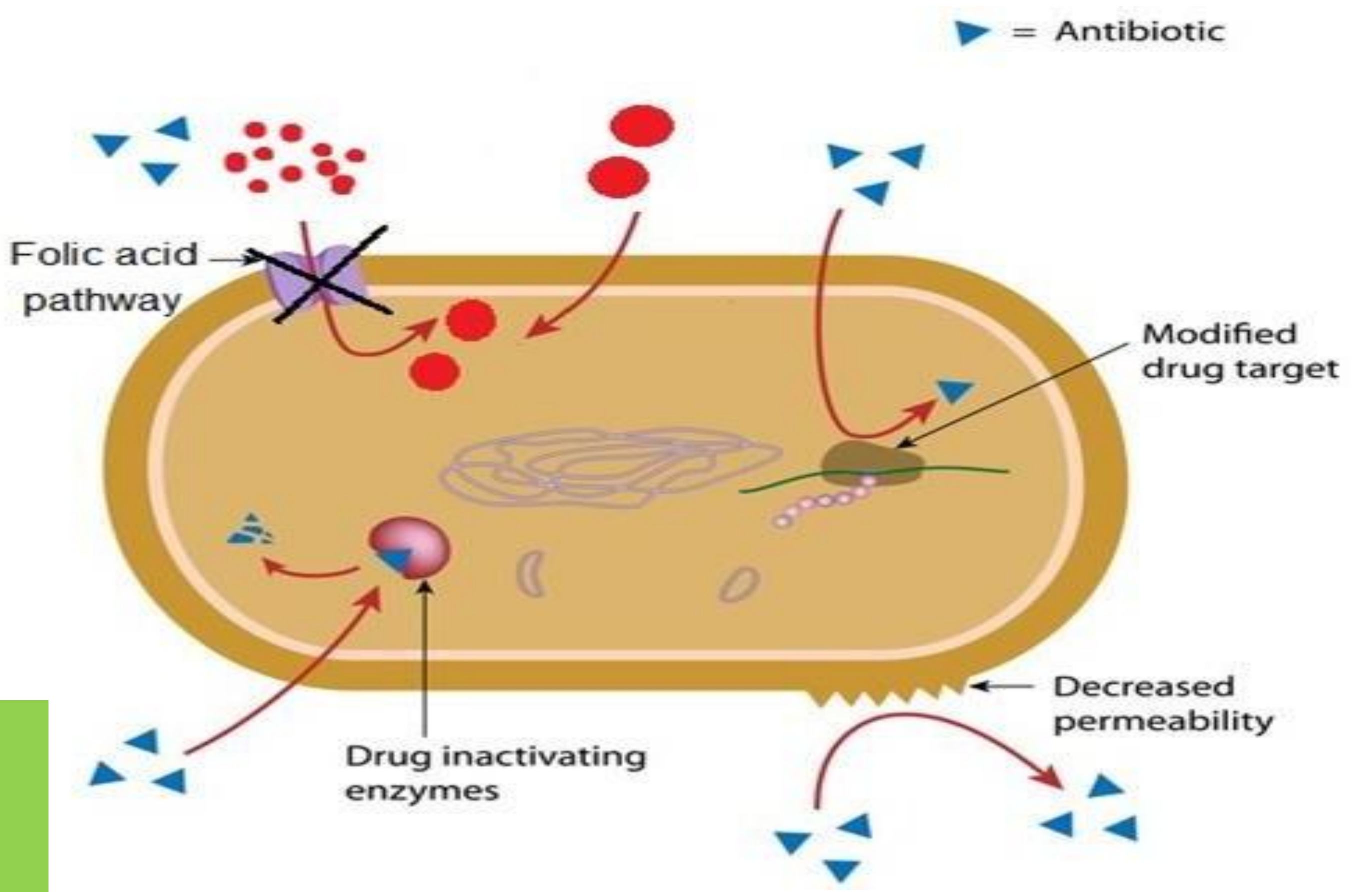
5. Differentiate between susceptible and resistant bacteria.

6. Appreciate the importance of sensitivity testing in preventing drug resistance.

Mechanisms of antimicrobial drug resistance

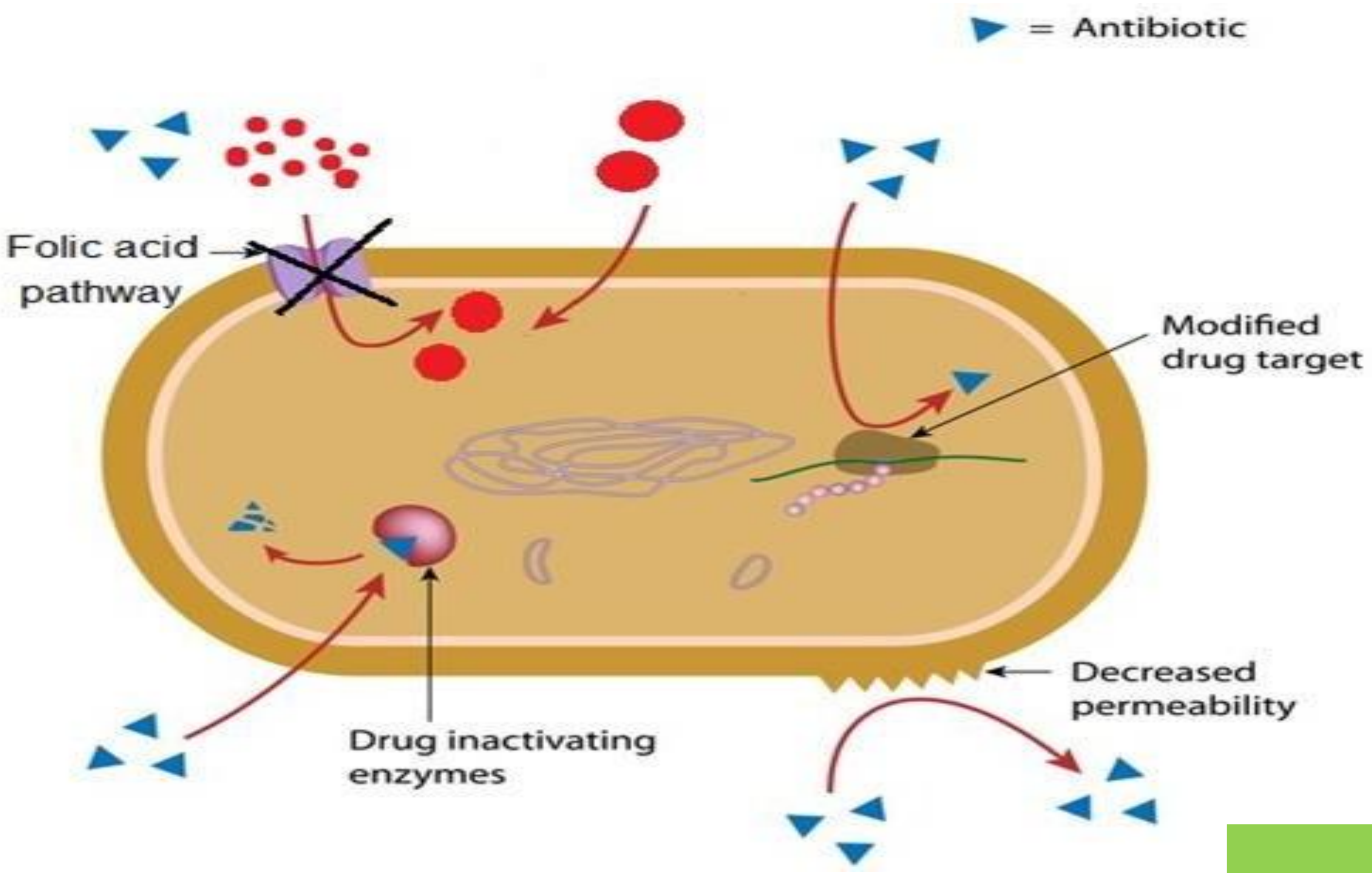


= Antibiotic
4
Folic acid pathway
3
Modified drug target
1
Drug inactivating enzymes
2
Decreased permeability



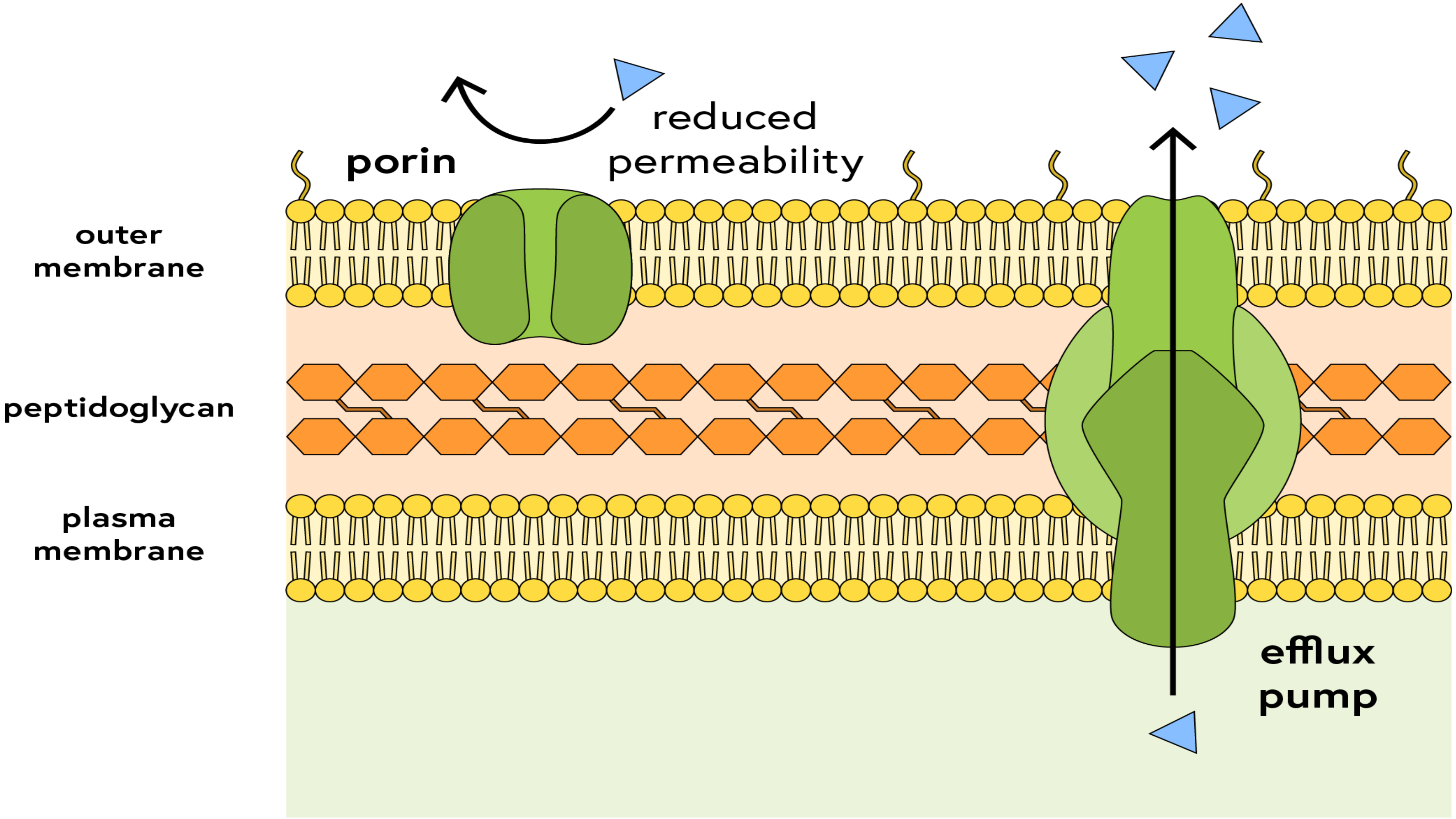
1

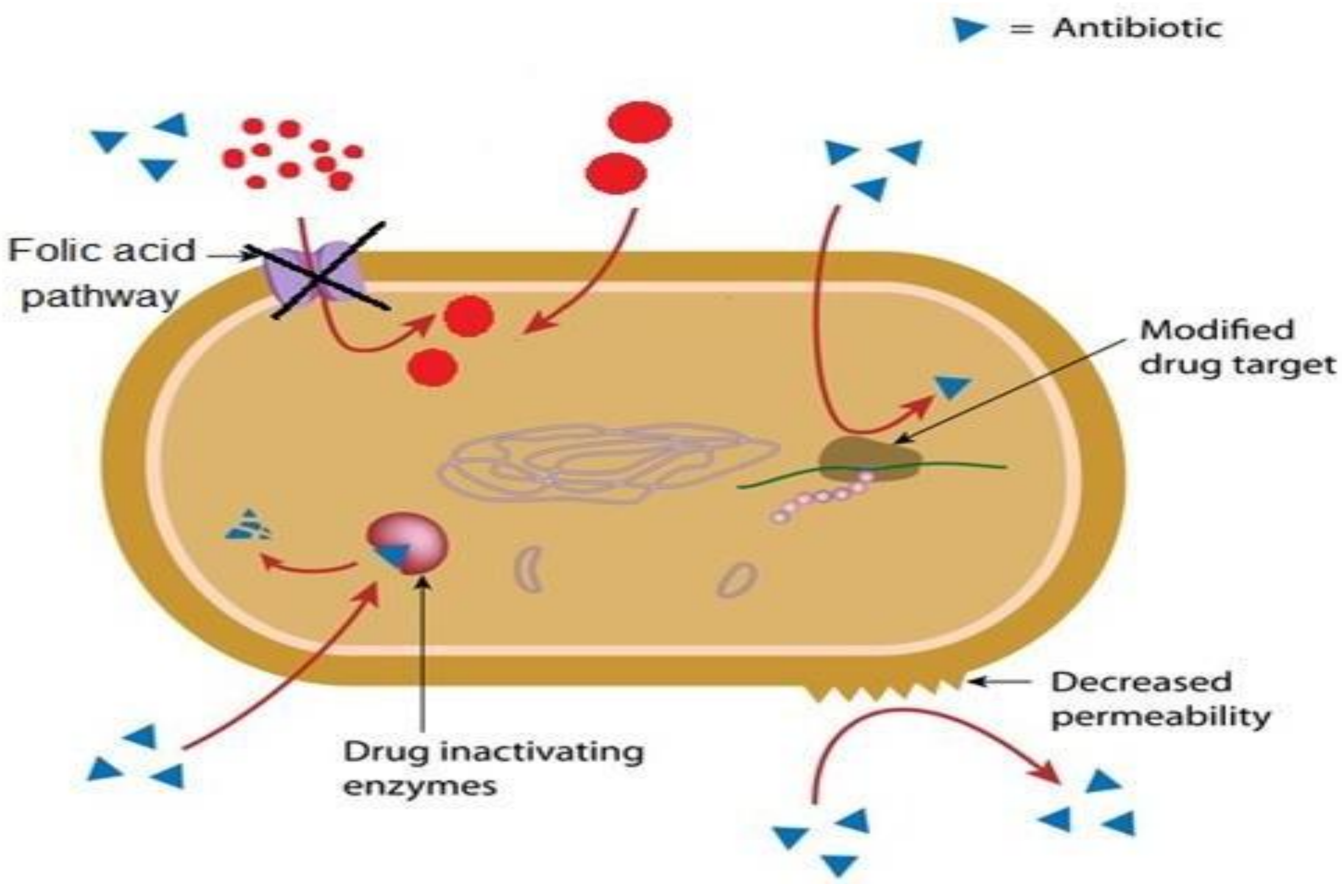
- Production of β- lactamase → destroy β- lactams
- Production of enzymes → destroys Aminoglycosides.



2

Decrease drug permeability→ no accumulation of drug inside the bacterial cell.(Tetracycline)


reduced
permeability
porin
outer
membrane
peptidoglycan
plasma
membrane
efflux
pump



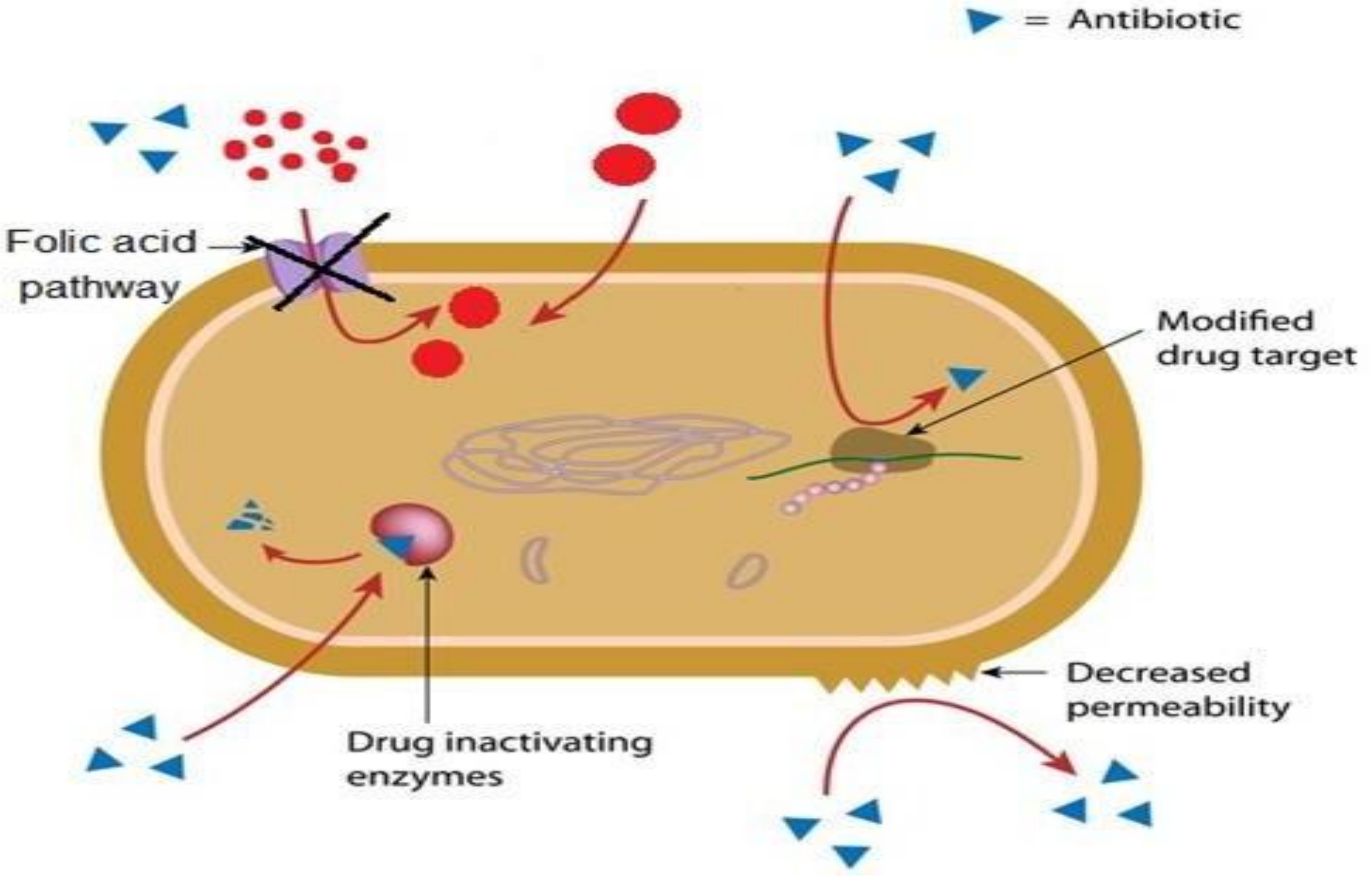
3

Altered receptor

- **On 30S subunit → resistance to Aminoglycosides and Tetracycline.**
- **on 50S subunit → resistance to Chloramphenicol and Erythromycin.**
- **Of PBPs → resistance to B-lactams**

4

- Altered metabolic pathway other than that inhibited by the drug
- Organisms depend on <u>preformed folic acid</u> other than PABA → resistance to Sulpha drugs.



4

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1

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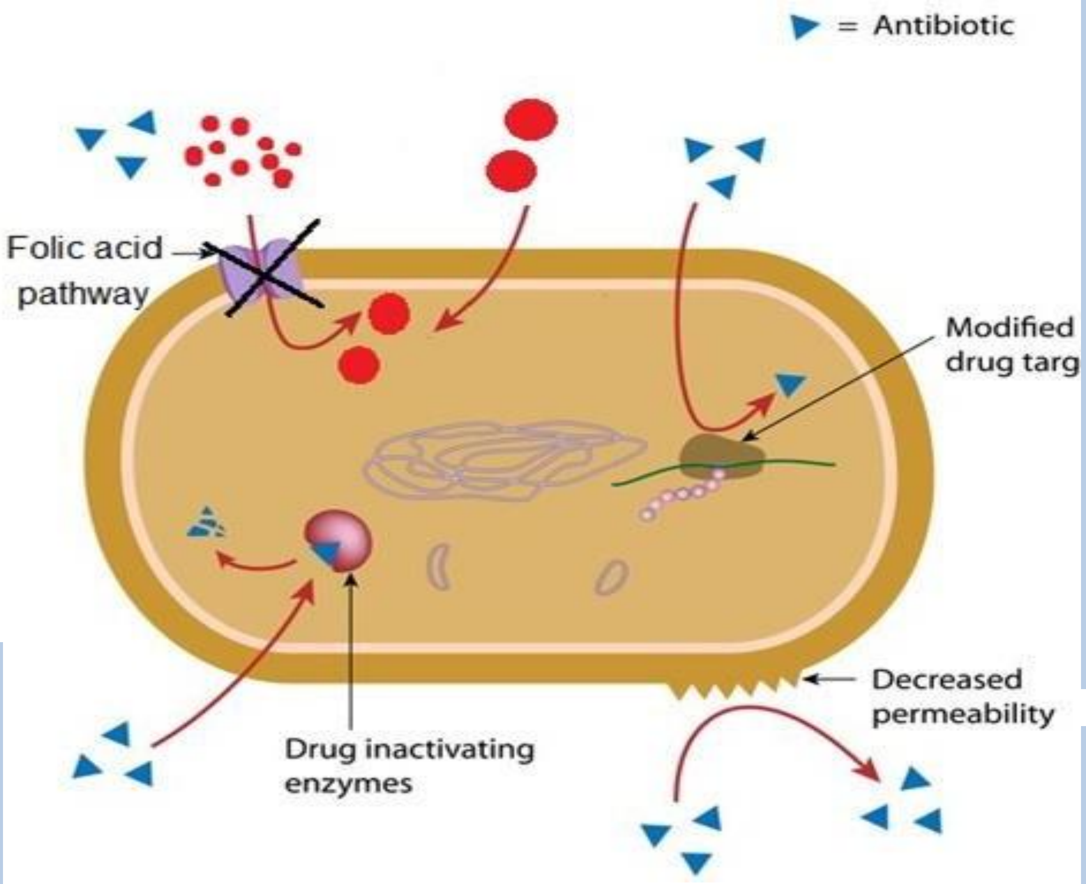

3

Altered receptor

- On 30S subunit → resistance to Aminoglycosides and Tetracycline.
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- Of PBPs → resistance to B-lactams

2

Decrease drug permeability→ no accumulation of drug inside the bacterial cell.(Tetracycline)

Origin of drug resistance

Non-Genetic

Resistance not caused by gene changes

1)Metabolic inactivity (dormancy) e.g. T.B

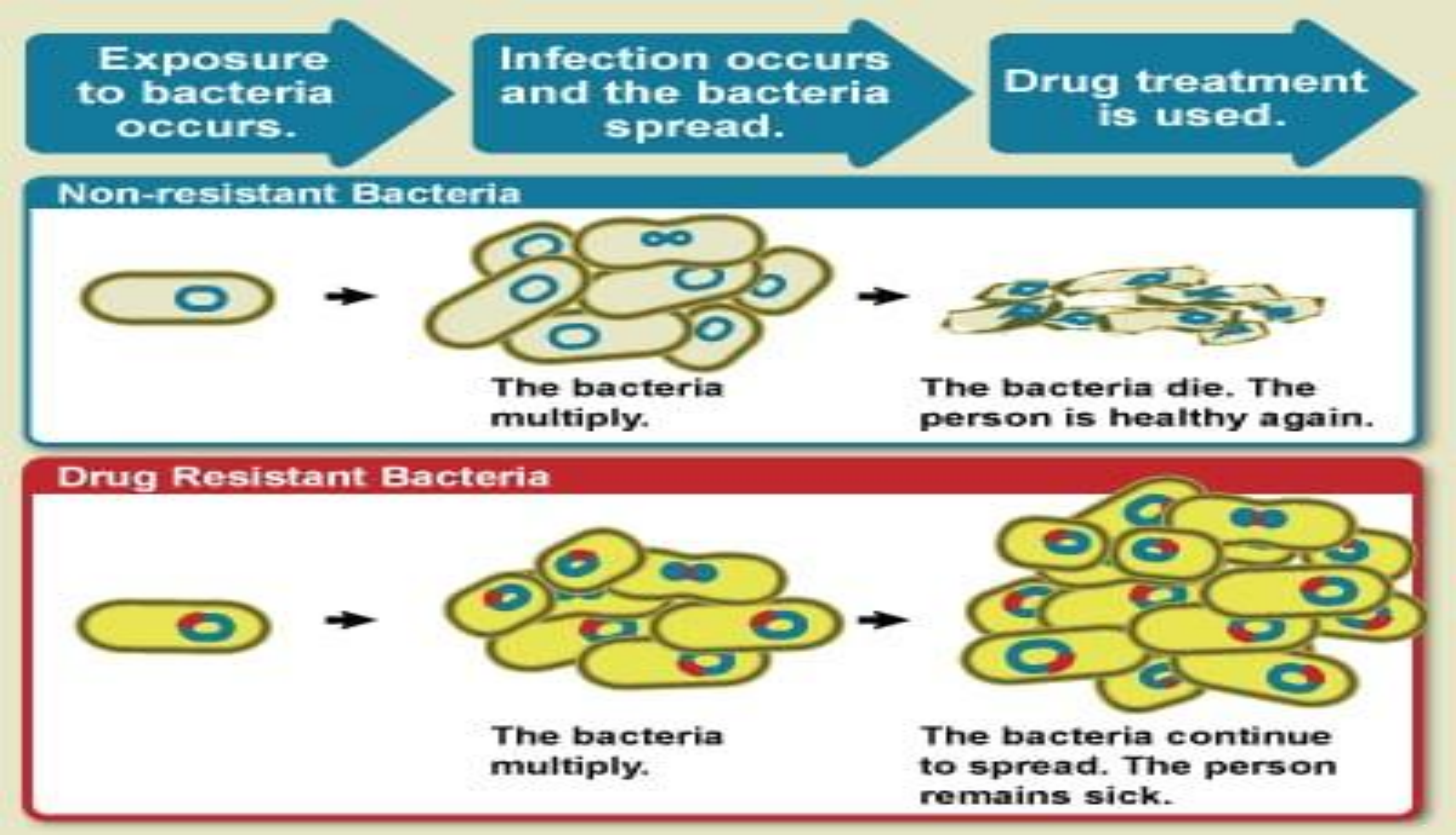
2)Loss of target structure e.g. lack cell wall in Mycoplasma Resistant to Penicillin

Genetic

Resistance due to changes in genes

1)Genetic mutation: e.g. change of PBPs

2)Gene transfer by transposons: plasmids or lactamase genes e.g. B-


Exposure to bacteria occurs.
Infection occurs and the bacteria spread.
Drug treatment is used.
Non-resistant Bacteria
The bacteria multiply.
The bacteria die. The person is healthy again.
Drug Resistant Bacteria
The bacteria multiply.
The bacteria continue to spread. The person remains sick.

ANTIBIOTIC SUSCEPTIBILITY TESTS

Testing a group of antibiotics against certain organism to choose the most effective on this organism. The most common methods are:

1. Disk diffusion test

bacteria is susciptible to **antibiotic A**

bacteria is Resistant to **antibiotic B**

The bacterial isolate is inoculated onto the surface of an agar Plate. A filter disk impregnated with a standard amount of an antibiotic is applied to the surface of the plate. The plates are incubated for 24 hr at 37°C. Following incubation, **zones of inhibition** of bacterial growth form around the **antibiotic disk**.

2. Broth dilution method:

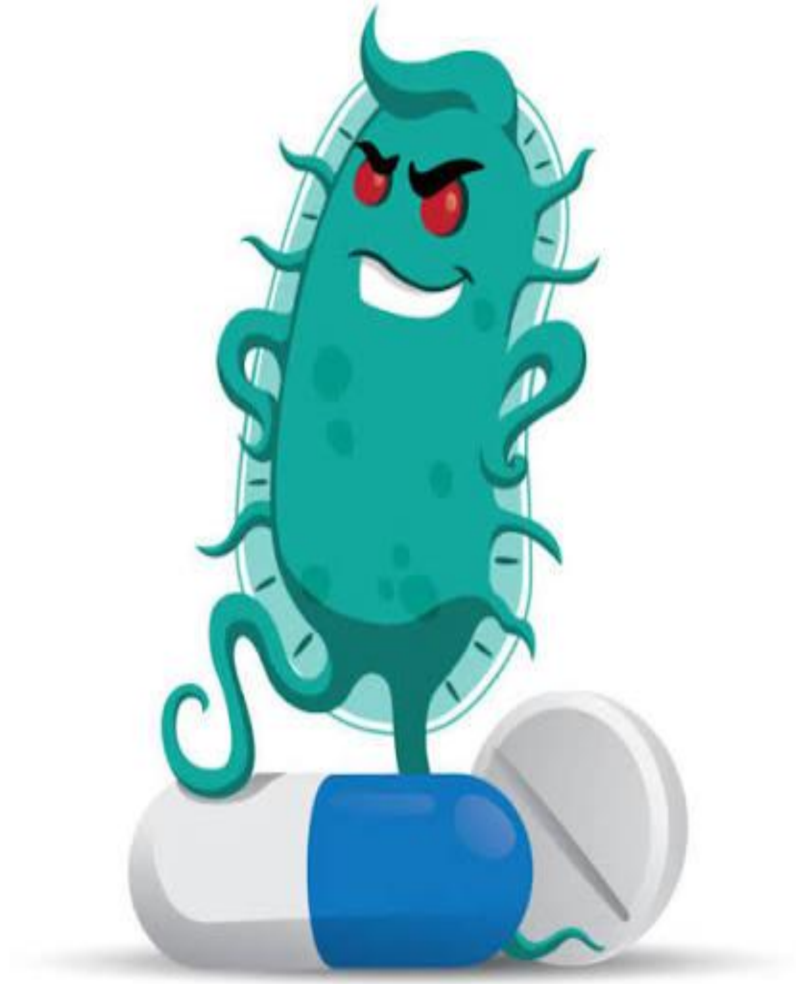
inoculating the organism in two fold dilutions of antimicrobial placed in tubes. After incubation at 35°C for 18 hours, we can measure **minimal inhibitory concentration** (MIC).

MIC: The MIC is the lowest concentration of the antibiotic that results in inhibition of bacterial growth.

Limitation of drug resistance

Avoid the usage of drug without (1
sensitivity test.

Maintain high levels of the drug in the (1
tissue throughout the course of the disease to reach the decline phase and prevent the resistance.



Complications of antibiotics

1. Drug resistance due to inadequate or prolonged use.

2. Drug toxicity specially in:

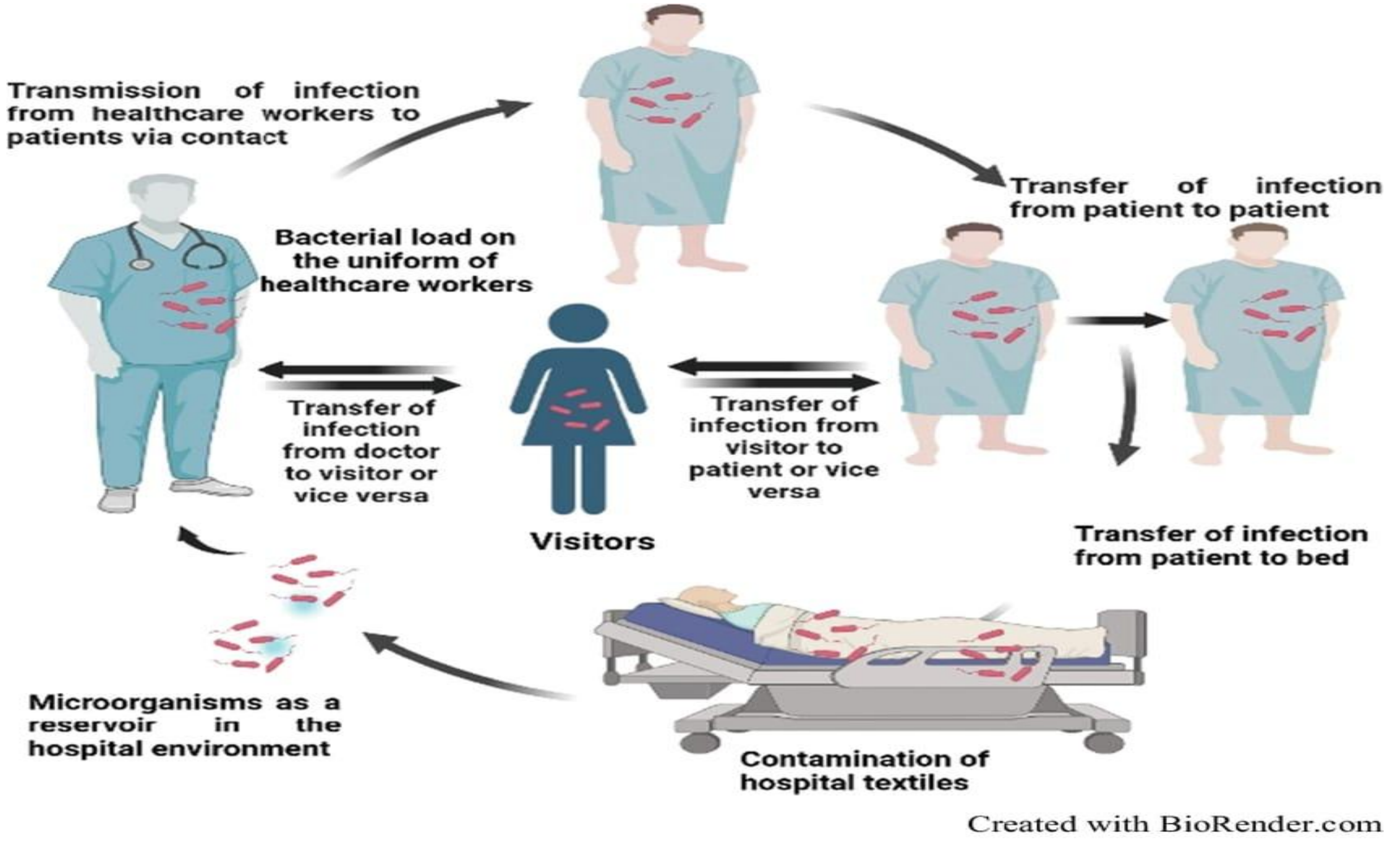
- Aminoglycosides → nephrotoxicity & ototoxicity
- Tetracycline → greyish coloration on the teeth
- Chloramphenicol → Agranulocytosis and aplastic anemia.

3. Allergic reaction in hypersensitive persons.

4. Early use of drug → suppress formation of antibodies → decrease natural defense → relapses of the disease.

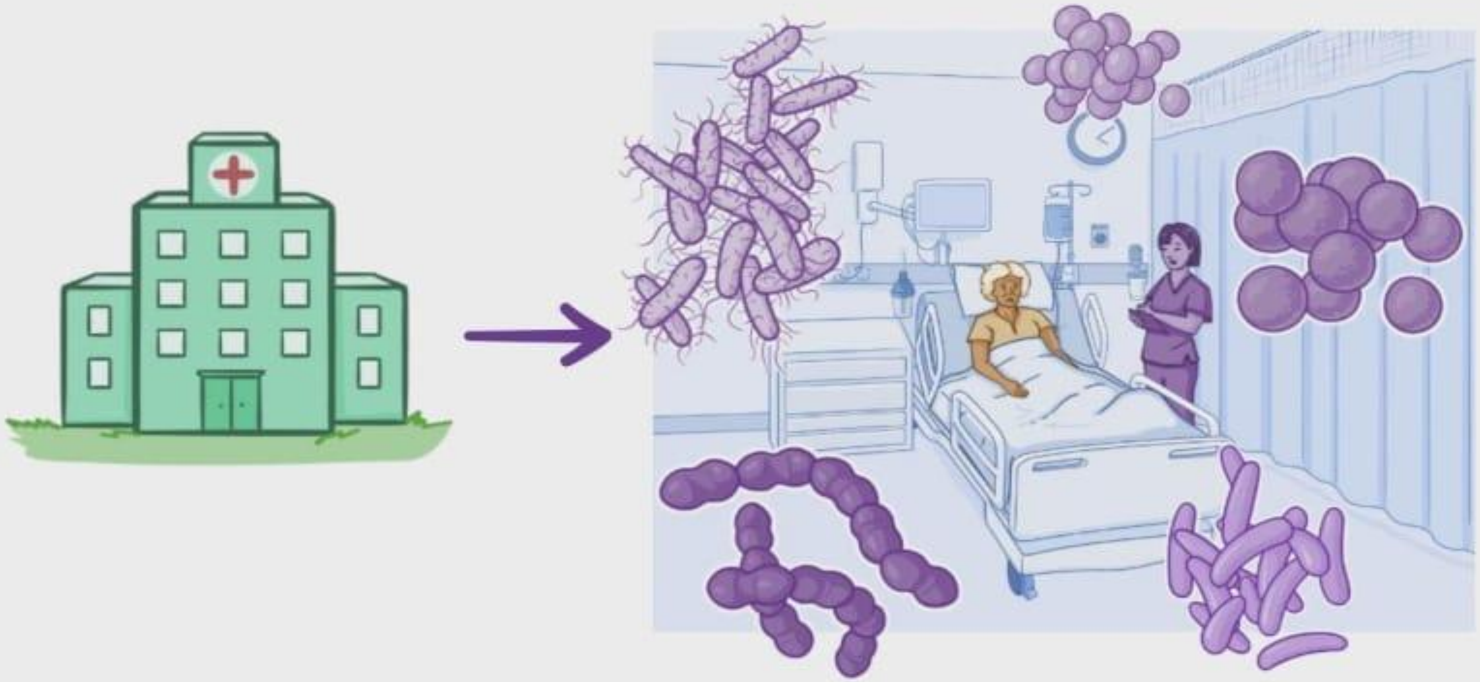
5. Prolonged use → superinfection (suppress of bacterial flora) as: Candida species


CAUSES OF
ANTIBIOTIC RESISTANCE
HANDLE ANTIBIOTICS WITH CARE
Antibiotic resistance happens when bacteria change and become resistant to the antibiotics used to treat the infections they cause.
Over-prescribing of antibiotics
Patients not finishing their treatment
Over-use of antibiotics in livestock and fish farming
Poor infection control in hospitals and clinics
Lack of hygiene and poor sanitation
Lack of new antibiotics being developed
www.who.int/drugresistance
#AntibioticResistance
World Health Organization


Transmission of infection from healthcare workers to patients via contact
Transfer of infection from patient to patient
Bacterial load on the uniform of healthcare workers
Transfer of infection from doctor to visitor or vice versa
Transfer of infection from visitor to patient or vice versa
Visitors
Transfer of infection from patient to bed
Microorganisms as a reservoir in the hospital environment
Contamination of hospital textiles
Created with BioRender.com

Nosocomial Infections

HEALTHCARE-ASSOCIATED INFECTIONS (HAIs)



- ~ HOSPITALIZED
- ~ LONG-TERM CARE
- ~ OUTPATIENT FACILITY


QUESTIONS

1-Which mechanism is responsible for pumping antibiotics out of the bacterial cell?

A. Porin channel

B. Efflux pump

C. Ribosomal mutation

D. Enzyme degradation

2-Which of the following is an example of genetic resistance?

A. Dormancy

B. Loss of target structure

C. Gene mutation

D. Reduced metabolism

3-A bacterium modifies its drug target so the antibiotic cannot bind. This is called:

A. Drug inactivation

B. Efflux

C. Altered receptor

D. Decreased permeability

3-In disk diffusion test, sensitivity to antibiotic is indicated by:

A. Presence of colonies

B. Color change

C. Zone of inhibition

D. Increased turbidity

4-Prolonged use of antibiotics may lead to:

A. Immunity enhancement

B. Superinfection

C. Reduced infection

D. Faster healing


THANK YOU